YUVRAJ GOVIND RAO

Greater Noida, UP 201310 | +91 7004890027 | yuvrajgovindrao@gmail.com | GitHub | Portfolio

SUMMARY

Computer Science graduate (2026), having hands-on experience building AI software in Python, built multi-tier application with FastAPI, React, REST API, PostgreSQL and Docker, built GraphRAG and RAG system using NLP (semantic search, vector embedding, entity extraction). Built data pipelines that integrate, cleanse and shape data into open-source database. Works with Jupyter Notebooks and foundational scikit-learn and TensorFlow. Tests with pytest and GitHub Actions CI. Projects are public on GitHub.

CORE SKILLS

Language: Python, JavaScript, TypeScript, SQL

ML and Data Science: AI/ML/DL fundamentals, Machine Learning (ML) frameworks: scikit-learn, Deep Learning (DL) frameworks: TensorFlow, Jupyter Notebooks

AI and NLP: Natural Language Processing (NLP), Retrieval Augmented Generation (RAG), GraphRAG, LLM API integration, semantic search, vector search, vector embedding, knowledge graph, entity extraction and resolution

Data and Database: PostgreSQL, SQLite, database design, data pipeline and dataflow

Backend and REST APIs: REST API design, Fast API, Express, Node.js, asyncio

Tools and DevOps: Git, GitHub, GitHub Actions (CI), pytest, Docker, Docker Compose, Azure Container Apps, debugging

EXPERIENCE

User Trial Program Participant - Nothing Technology *2024 - 2025*

- Documented 30+ structure defect report for bugs and performance issue, many critical, across 2 unreleased devices using the program feedback framework that directly informed product decision ahead of launch.

PROJECTS

GraphRAG - Knowledge Graph Q&A Engine

Python, FastAPI, Qdrant, Neo4j, Google Gemini, OpenAI | https://github.com/yuvrajgovindrao/GraphRAG

- Built a Retrieval Augmented Generation (RAG) microservice using Python/FastAPI combining Qdrant for vector search with Neo4j for graph traversal to produce cited, multi document Q/A.
- Build an automated entity and relationship extraction pipeline using UNWIND batched Cypher query, in under 1.5 seconds merging 100+ entities into Node4j.
- Rebuild the graph expansion logic to resolve a cross-document retrieval gap, removed broken relationship links between linked entities across sourced documents.

YouTubeRAG - Multi-Video RAG Assistant

Python, FastAPI, PostgreSQL + pgvector, React/Vite, Gemini | https://github.com/yuvrajgovindrao/YouTubeRAG

- Built a full stack RAG application using React/Vite, FastAPI, PostgreSQL + pgvector that indexes and ingest YouTube videos and playlist, provides cross video Q/A with seconds accurate clickable timestamp grounded citations.
- Wrote SQL based retrieval logic in PostgreSQL using pgvector to store and query 768 dimension Gemini embeddings via cosine-similarity search.
- Architected a single config driven Docker Compose codebase supporting both unrestricted local development and a rate limited public demo (10 requests/hour), with GitHub Actions CI running the test suite on every commit.

TeamFlow: Team Task Manager

React, Node.js, Express, PostgreSQL | https://github.com/yuvrajgovindrao/team-task-manager

- Build an authentication system using bcrypt hashed password, OTP signup and JWT session, including a password reset feature from scratch.
- Implements admin/member roles for every project within the Express middleware, where multi-assignee tasks are modelled via a junction table across four versioned SQL migrations.

macOS-Inspired Interactive Portfolio

React 19, TypeScript, Zustand, GSAP, Gemini | https://github.com/yuvrajgovindrao/macos-portfolio

- Built a desktop-style portfolio using React 19 and TypeScript, featuring a Zustand-based window manager that handles the open/close/focus/drag states of over five simulated windows; also includes GSAP animations and a responsive dark-mode/iOS-style mobile layout.
- Integrated a Google Gemini-powered “Ask AI” assistant so visitors can ask questions about background and projects.

EDUCATION

Bachelor of Technology in Computer Science and Engineering, Galgotias University, 2026

CERTIFICATIONS

Google Android Developer Certification – AICTE | AWS Data Engineer - AICTE